

PROJECT DOCUMENTATION

Auto Ticket Classification using Flow Designer

Project title	Auto Ticket Classification using Flow Designer
Technology/platform	ServiceNow – Flow designer
Project type	Automation Project
Repository	Public GitHub Repository – Auto Ticket Classification using Flow Designer
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Institution / department	St.Joseph's Arts and Science College for Womens
Programme/year	B.Sc.Computer Science -3 rd year
Submission date	25-09-2026

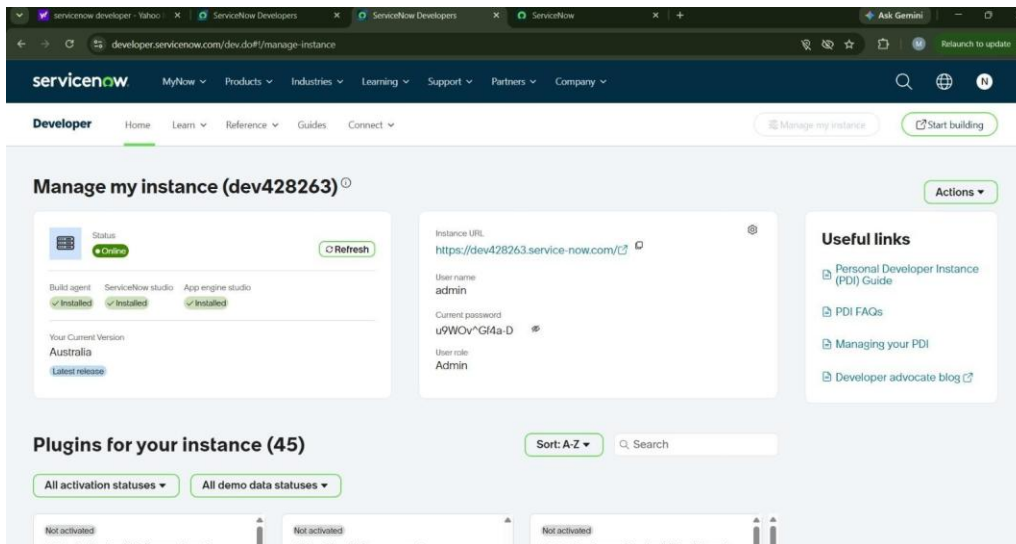
Abstract

Auto Ticket Classification using Flow Designer is a system designed to automatically classify and manage support tickets based on their details. In a traditional ticketing system, tickets are manually reviewed and assigned to the appropriate category or support team, which can take time and may cause delays. This project uses Flow Designer to automate the ticket classification and assignment process.

Objective

- Automatically classify incidents at the time of creation.
- Reduce manual effort for IT agents.

- Improve ticket routing efficiency.
- Implement a no-code, easily maintainable solution.



PHASE 1: Requirement Analysis & Planning

The school IT helpdesk receives multiple support requests daily from students and teachers related to common issues such as network connectivity, hardware failures, account access problems, and system performance issues.

Currently, IT staff manually review each ticket to:

- Identify the issue type.
- Select the correct category and subcategory.
- Inform the caller about ticket creation.

This manual process is time-consuming, error-prone, and not scalable as the number of requests increases.

Activity : 1 Business Use Case.

Business Requirements

The system must:

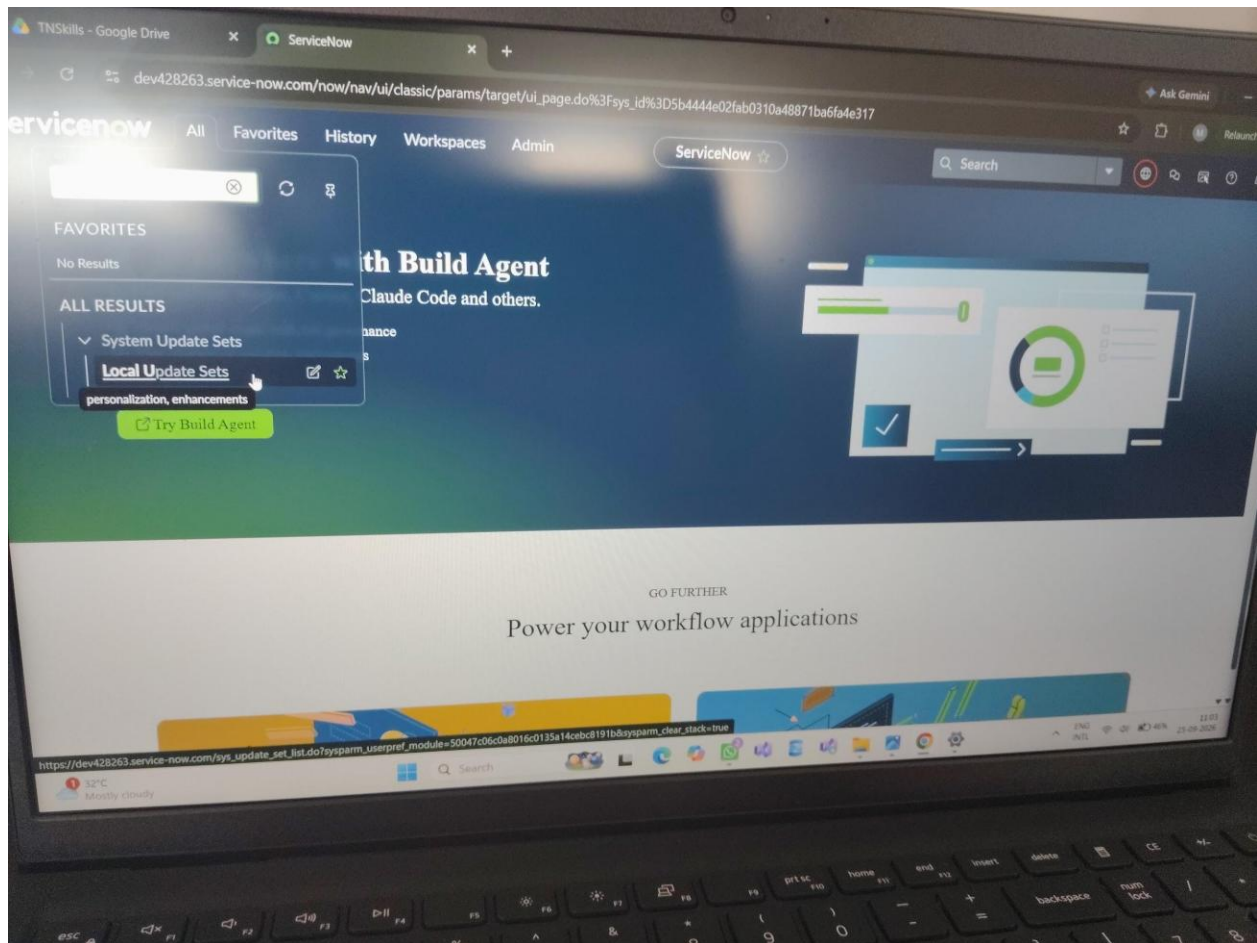
1. Automatically classify IT tickets based on the issue description.
2. Assign both Category and Subcategory without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller upon ticket creation.
5. Store ticket information in a structured and standardized format.
6. Ensure easy maintenance and future scalability.

Activity : 2 Creation of New Update Set.

Navigation Steps :

- Go to All menus → Search for Update set.
- Under System Update Sets → Select Local Update Sets.
- Click the New button on the top right corner of the page.
- Name : **Project Update Set.**
- State : **In progress.**
- Application : **Global.**
- Then **Save** the **Form.**
- Under Related links Click on the **Make This My Current Set.**
- Then the Created Update Set will appear as Current.

Activity : 3 Navigation steps in Picture View.



Update Sets							
Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
Default	Global	In progress		2026-06-12 14:20:10	system	(empty)	(empty)
Project Update Set	Global	In progress		2026-09-24 21:43:59	admin	(empty)	(empty)
Project Update Set	Global	Complete		2026-09-22 07:11:02	admin	(empty)	(empty)
Project Update Set 2	Global	In progress		2026-09-24 21:05:39	admin	(empty)	(empty)

Related Links
Merge Update Sets

1 to 4 of 4

Phase :- 2 Backend Development & Configuration

Activity 1 : Creation of Custom Table to Store Ticket Records.

Steps to Create Custom Table

- Open All menus → Search for Tables.
- Under **System Definition** → Click **Tables**.
- On the top right corner Click **New** button.
- Label : **Incident WorkFlow**.
- Uncheck the **Create module**.
- Now **Save** the Form.
- Then Click the **Controls** Suction and Enable the Check box **true Auto Number**.
- Now again **save** the Form.

The screenshot shows the ServiceNow interface for creating a new record in an update set. The browser address bar shows the URL: `dev428263.service-now.com/now/nav/ui/classic/params/target/sys_update_set.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_update_set%26sysparm_chec...`. The page title is "Update Set - Create Project Update Set 2". The form fields are as follows:

- Name:** Project Update Set
- State:** In progress
- Parent:** (empty)
- Release date:** (empty)
- Description:** (empty)
- Application:** Global

At the bottom of the form, there are two buttons: "Submit" and "Submit and Make Current".

Table - Incident Workflow

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More info](#)

* Label: Incident Workflow
 * Name: u_incident_workflow
 Application: Global
 Remote Table:

Columns Controls Application Access

Extensible ☐
 Live feed ☐

Use auto-numbering to define a sequential identifying code made up of a prefix, a base number and a padding value to ensure a consistent format.

Prefix: INC
 Number: 1,000
 Number of digits: 7

Security Rules (ACLs) are required if anyone other than an administrator needs to work with this table. Creating default security rules will grant full access to this table to anyone with the user role you specify.

Create access controls ☒

* User role: u_incident_workflow_user

- Then Save the Form and Under Related Links Click **Show Form**.
- Now Open the Configure On clicking the Context Menu and then Open the **Form Designer**.
- Then Create Fields as Required.

Incident Workflow

Save
 Insert
 Insert and Stay
 Analyze Access
 Configure
 Export
 View
 Create Favorite
 Copy URL
 Copy sys_id
 Show XML
 History
 Reload form

Form Builder
 Form Design
 Form Layout
 Related Lists
 All
 Table
 Security Rules
 Business Rules
 Client Scripts
 UI Policies
 Data Policies
 UI Actions
 Notifications
 Dictionary

Number: INC00519
 * Caller: Janice Twiet
 State: -- None --
 Assigned group
 Assigned to

Update Delete

Table - Incident Workflow

Number of digits: 7

Security Rules (ACLs) are required if anyone other than an administrator needs to work with this table. Creating default security rules will grant full access to this table to anyone with the user role you specify.

Create access controls ☒

* User role: u_incident_workflow_user

Related Links: Form Builder, Design Form, Layout Form, Layout List, Show Form, Show List, Show Schema Map, Add to Service Catalog, Run Point Scan, Explore REST API

Access Controls (4) | Security Data Filters | Labels (1) | Database Indexes (1) | Table Subscription Configuration (1)

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_incident_workflow	Allow If	read	record	true	admin	2026-09-22 07:19:59

Activity 2: Field Creation and Data Type Configuration

The objective of this milestone is to create all required fields in the School IT Ticket table with correct data types, choices, and dependencies, ensuring structured and consistent data capture.

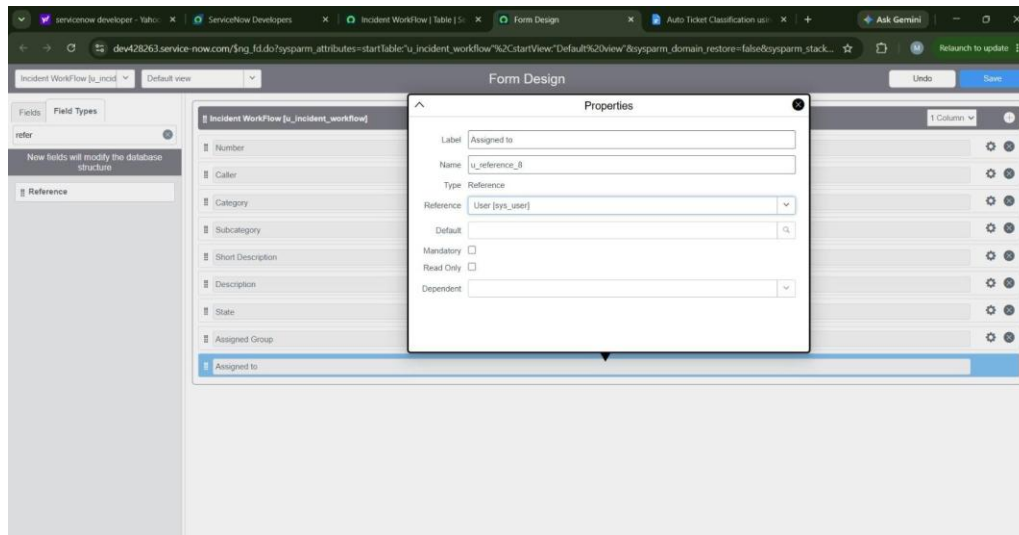
Field Label	Type	Reference/Choice
Number	Auto Number	—
Caller	Reference	sys_user
Category	Choice	[Network,Hardware,Access,Performance]
Subcategory	Choice	[Wi-Fi,Projector,Forgot Password,Slow Computer]
Short Description	String	—
Description	String	—
State	Choice	[New,In progress,On hold,Resolved,Closed]
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

- After Creating Fields with their appropriate Type, then add the Choice from the Form design.
- To all the Choice list Fields.
- They are Category, Subcategory and State.

The screenshot shows the ServiceNow Form Design interface for the 'Incident Workflow' form. The left sidebar has a 'Fields' section with a list of fields: 'Created', 'Created by', 'Updated', 'Updated by', and 'Updates'. Below this is a 'Formatters' section with 'Activities (filtered)', 'Contextual Search Results', and 'Ratings'. The main area shows a single field named 'Number'.

The screenshot shows the ServiceNow Form Design interface for the 'Incident Workflow' form. The left sidebar has a 'Fields' section with a list of fields: 'Created', 'Created by', 'Updated', 'Updated by', and 'Updates'. Below this is a 'Formatters' section with 'Activities (filtered)', 'Contextual Search Results', and 'Ratings'. The main area shows a single field named 'Number'. A 'Properties' dialog box is open, showing the configuration for the 'Caller' field. The dialog box has the following fields: 'Label' (Caller), 'Name' (u_Caller), 'Type' (Reference), 'Reference' (User [sys_user]), 'Default' (empty), 'Mandatory' (checkbox), 'Read Only' (checkbox), and 'Dependent' (empty).

- Same like that, add Choice to the State field.
- Then Click the save button on the top right corner.
- Open the Form and reload it.



Activity 3 : Implementing Dependency Between Category and Subcategory Choice Fields

The objective of this milestone is to implement a **dependency relationship** between the **Category** and **Subcategory** fields so that the available Subcategory options dynamically change based on the selected Category.

Navigation Steps :

- Open the Created Table Form and Right click on the **Subcategory** Field.
- Click on the **Configure Dictionary**.
- Under **Related links** Click on the **Advanced view**,
- Then select the Dependent **Field** Section.
- Enable the checkbox **Use dependent field true**.
- Then select the Field **Category** on the **Dependent on field**.
- Then **save** the Form.

servicenow Dictionary Entry - Subcategory

Displays a list of suggested values in a Choice list. In the Advanced view you can select the Choice table and the Choice field to take choice values from, plus a Dependent field.

Choice Dropdown with -- None --

Create Choice List Delete Column Update

Related Links

Show Table
Run Point Scan
Advanced view

Access Controls Choices (4) Attributes (1) Labels (1)

Name Search

Access Controls

Name	Decision Type	Operation	Type	Active	Updated by	Updated
						
No records to display						

servicenow Dictionary Entry - Subcategory

Attributes edge_encryption_enabled=true

Choice List Specification Dependent Field Calculated Value Default Value

Specify a field whose value is used to determine choice list options for this field. [More Info](#)

Use dependent field ☒

Dependent on field Category

Create Choice List Delete Column Update

Related Links

Show Table
Run Point Scan
Default view

Access Controls Choices (4) Attributes (1) Labels (1)

Name Search

Access Controls

Name	Decision Type	Operation	Type	Active	Updated by	Updated
------	---------------	-----------	------	--------	------------	---------

- Save
- Insert
- Insert and Stay
- Analyze Access
- Show File Properties
- Move to Application...
- Show Latest Update
- Configure
- Export
- View
- Create Favorite
- Copy URL
- Copy sys_id
- Show XML
- History
- Reload form

- Now scroll down to the choice section.
- Then add the dependent fields for which option dependent on the category gives the category choice backend name to the subcategory choice values.
- Wi-Fi → Network.
- Projector → Hardware.
- Forgot password → Access.
- Slow computer → Performance.

Dictionary Entry - Subcategory

Use dependent field ☒

Dependent on field: Category

Related Links: Show Table, Run Point Scan

Access Controls: Choices (4), Attributes (1), Labels (1)

Label	Value	Language	Sequence	Inactive	Updated	Dependent value
Slow Computer	slow computer	en	3	false	2026-09-22 07:49:40	Performance
Projector	projector	en	1	false	2026-09-22 07:49:40	Network
Wi-Fi	wi-fi	en	0	false	2026-09-22 07:49:40	Access
Forgot Password	forgot password	en	2	false	2026-09-22 07:49:40	

- Then save the Form.
- Then it will display each choice to each one separately based on the category subcategory displays.

Phase 3 : Automation using Flow Designer & Email Notification.

When a student creates a new IT support ticket, they only need to enter the Caller Name and Short Description of the issue. The student does not need to manually select the Category or Subcategory. Once the ticket is submitted, a Flow Designer automation is triggered and evaluates the Short Description for predefined keywords such as WiFi, Projector, Password, or Slow. Based on the identified keyword, the flow automatically assigns the appropriate Category and Subcategory to the ticket.

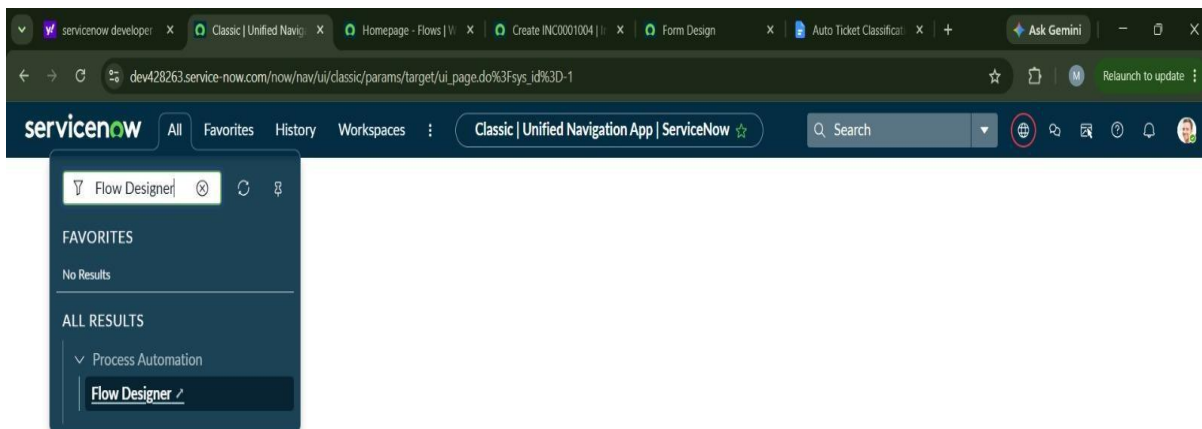
- If the short description contains the word *WiFi*, the flow assigns **Category = Network** and **Subcategory = WiFi**.
- If the description contains *Projector*, the flow assigns **Category = Hardware** and **Subcategory = Projector**.
- If the description includes *Password* or *Login*, the flow assigns **Category = Access** and **Subcategory = Forgot Password**.
- If the description mentions *Slow* or *Hanging*, the flow assigns **Category = Performance** and **Subcategory = Slow Computer**.

This automation ensures consistent classification, eliminates human error, and significantly reduces the manual effort required by IT staff.

The entire process is implemented using **Flow Designer**, making the solution easy to maintain, scalable, and completely no-code.

Activity : 1 Auto Classification using Flow design.

- Go to **All menus** → Search **Flow designer**.
- Under **Process Automation** Open the **Flow designer**.
- On the top right corner Click **New** and Select **Flow** Option.
- Name : **Auto Classify School IT Tickets**.
- Application : **Global**.
- Now Click the **Build Flow** button.



Let's get the details for your flow

Flow name

Application

Description

[Show additional properties](#)

Activity : 2 Trigger Configuration.

- Trigger : **Record Created.**
- Table : **Incident Workflow.**
- Condition : **Category is Empty.**
- Then Click **Done.**

Auto Classify School IT Tickets inactive

TRIGGER

Incident Workflow Created where (Category is null)

Trigger:

* Table:

Condition: is or and

ACTIONS Select multiple

Add an Action, Flow Logic, or Subflow

Data Collapse All

Flow Variables

Trigger - Record Created

Incident Workflow Record	Record
Incident Workflow Table	Table
Run Start Time UTC	Date/Time
Run Start Date/Time	Date/Time

- Now Under **Actions** Select **Flow Logic** Option.

Activity : 3 Wi-Fi Issue Classification.

- Select **If** condition.
- Condition label : SD is **Wifi or Network** (Give as ur wish).
- Condition 1 : Using data pill picker Select **Short description** as Field , and operator **Contains** , Value is **Wi-Fi**.
- Condition 2 : Using data pill picker Select **Short description** as Field, and Operator is **Contains**, Value is **Network**.
- Then Click **Done** Button.
- Now Click **Action** to **Update record**.

The screenshot displays the ServiceNow Workflow Studio interface for a workflow named "Auto Classify School IT Tickets". The workflow is currently inactive. The configuration is as follows:

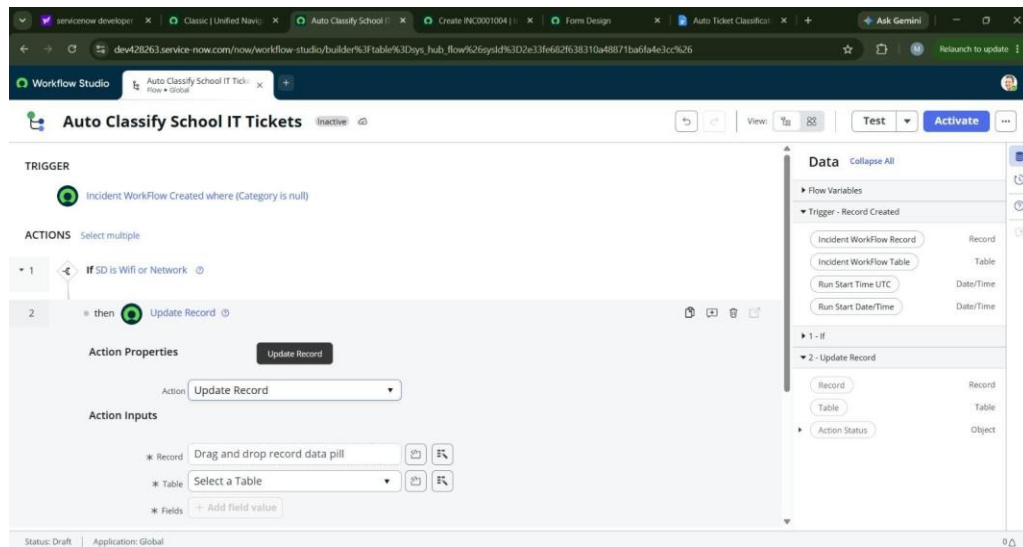
- TRIGGER:** Incident Workflow Created where (Category is null)
- ACTIONS:** Select multiple
 - 1 If**
 - Condition Label:** SD is Wifi or Network
 - Condition 1:** Trigger - ... Short Descri... is Wi-Fi
 - OR**
 - Condition 2:** Trigger - ... Short Descri... is Network
 - Add another condition set(OR)**
 - Buttons:** Delete, Cancel, Done

Data Panel (Right):

- Flow Variables:**
 - Incident Workflow Record (Record)
 - Incident Workflow Table (Table)
 - Run Start Time UTC (Date/Time)
 - Run Start Date/Time (Date/Time)
- 1 - If**

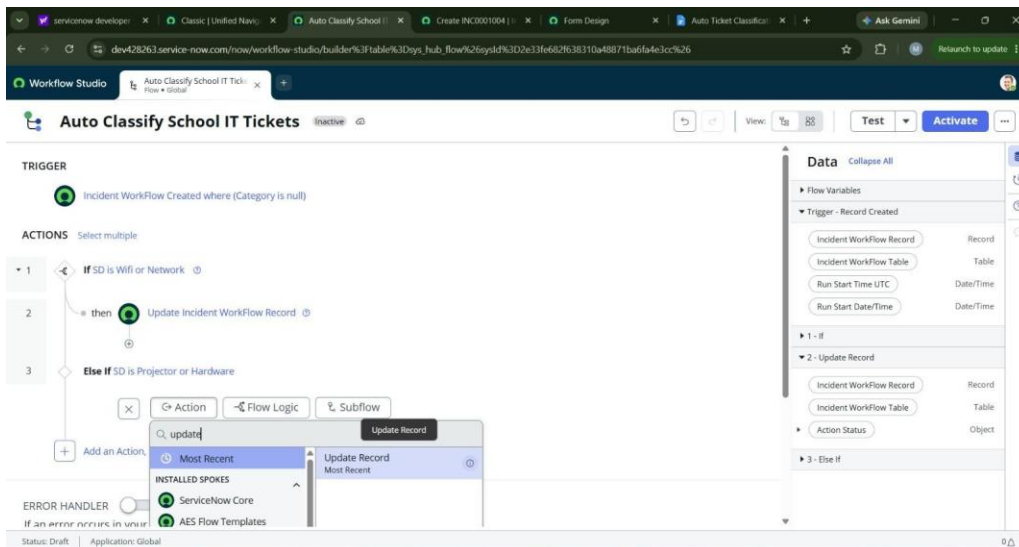
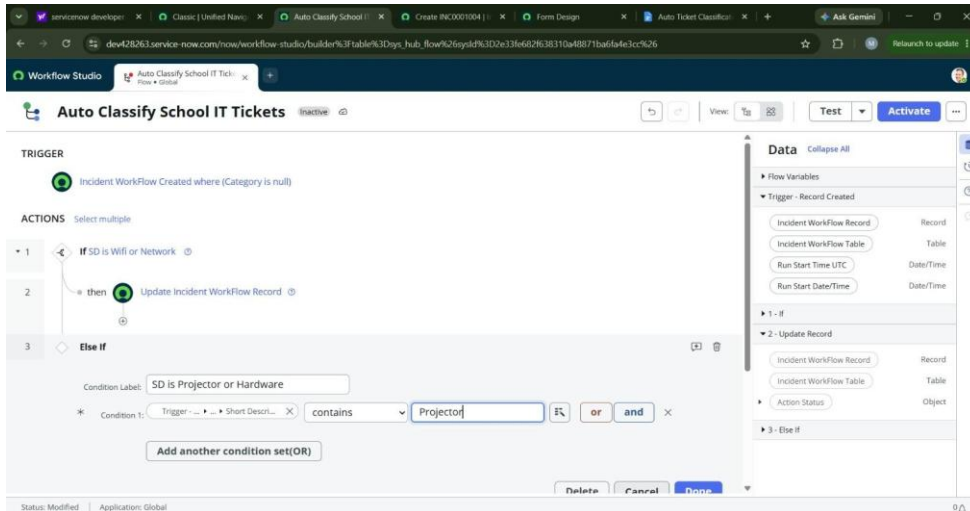
Status: Modified | **Application:** Global

- Action Inputs.
- In Record Field drag and drop data from the right side.
- Table : Auto updates as Incident Workflow.
- Fields Set **Category = Network**.
- Set **Subcategory = Wi-Fi**.
- Then Click the **Done** button.



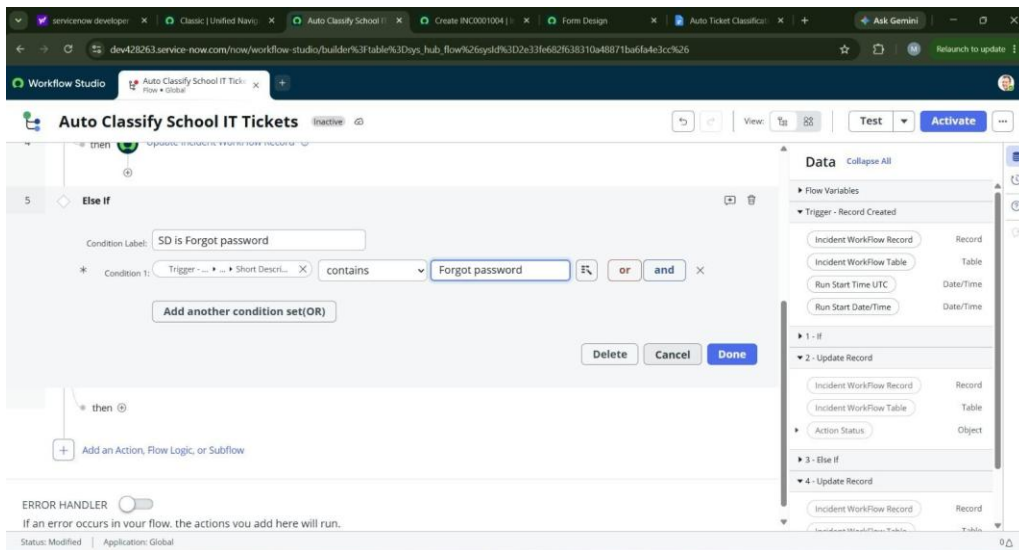
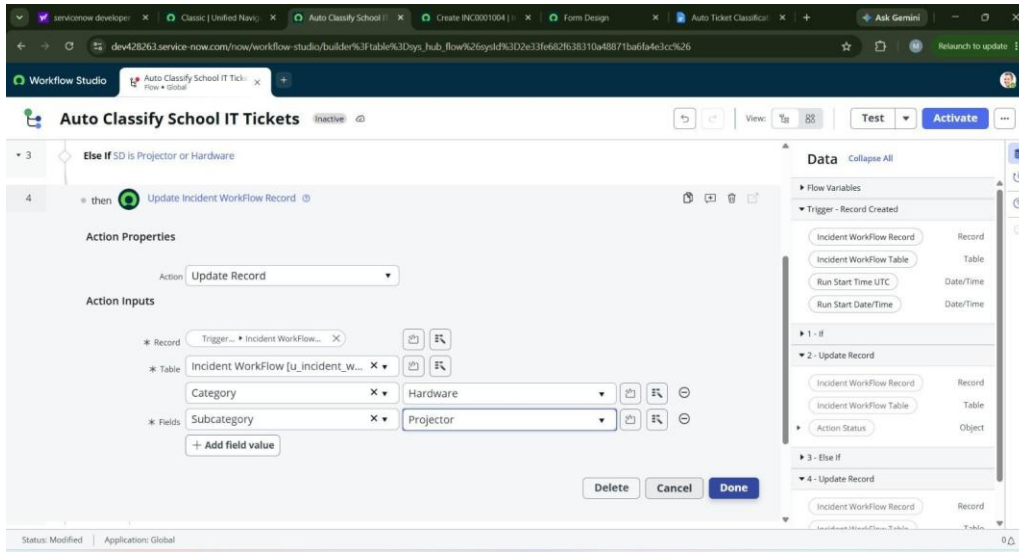
Activity : 4 Projector Issue Classification.

- Now Select the **Else If** Condition.
- Condition label : SD is **Projector or Hardware**.
- Condition : **Short description Contains Prajector**.
- Then Click the **Done** button.
- Now Action is an **Update record**.
- Set **Category = Hardware**.
- Set **Subcategory = Projector**.
- Then Click the **Done** button.



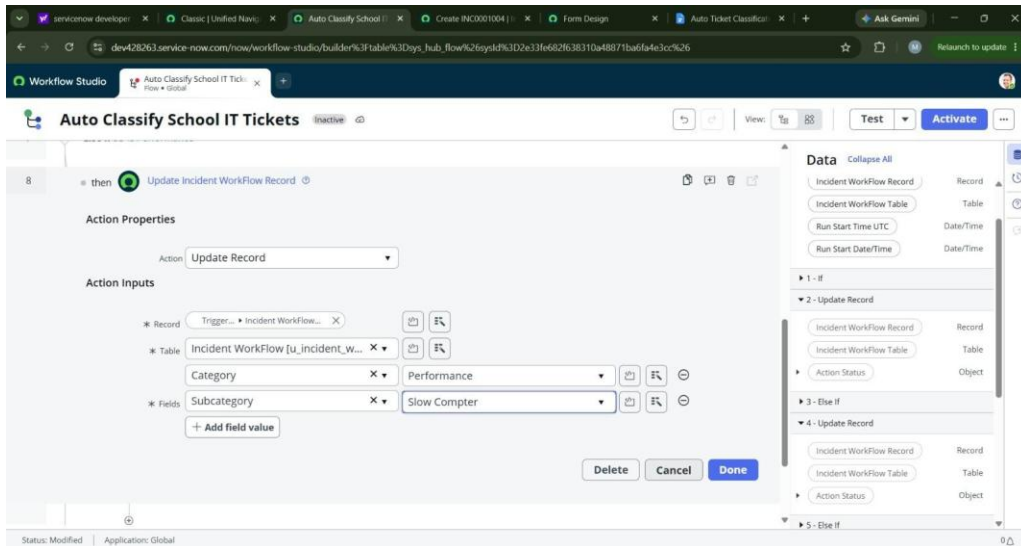
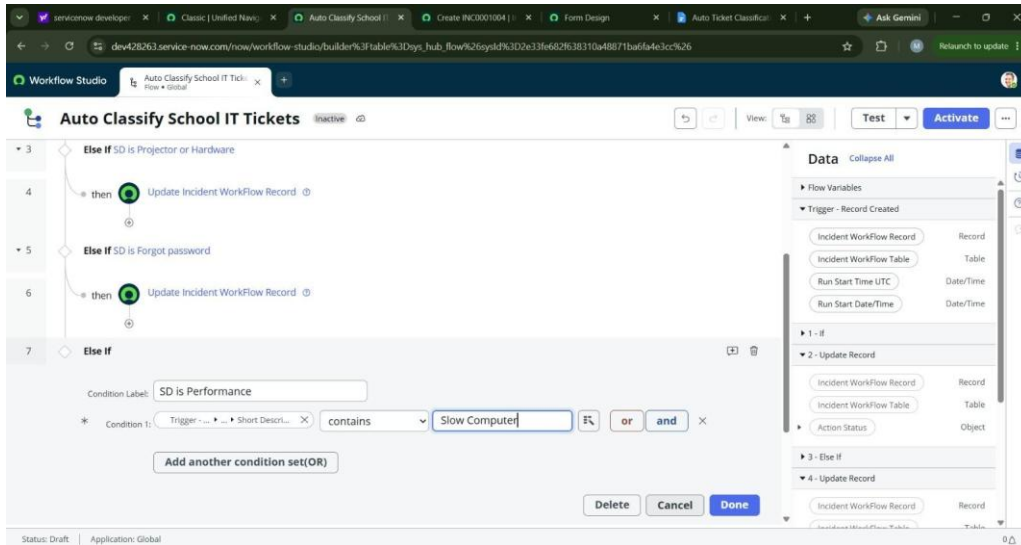
Activity : 5 Password Issue Classification.

- Now again select **Else If** condition.
- Condition label : SD is **Forgot password**.
- Condition : **Short description contains Forgot password**.
- Then Click the **Done** button.
- Now Action is an **Update Record**.
- Set **Category** = **Access**.
- Set **Subcategory** = **Forgot Password**.
- Then Click the **Done** button.



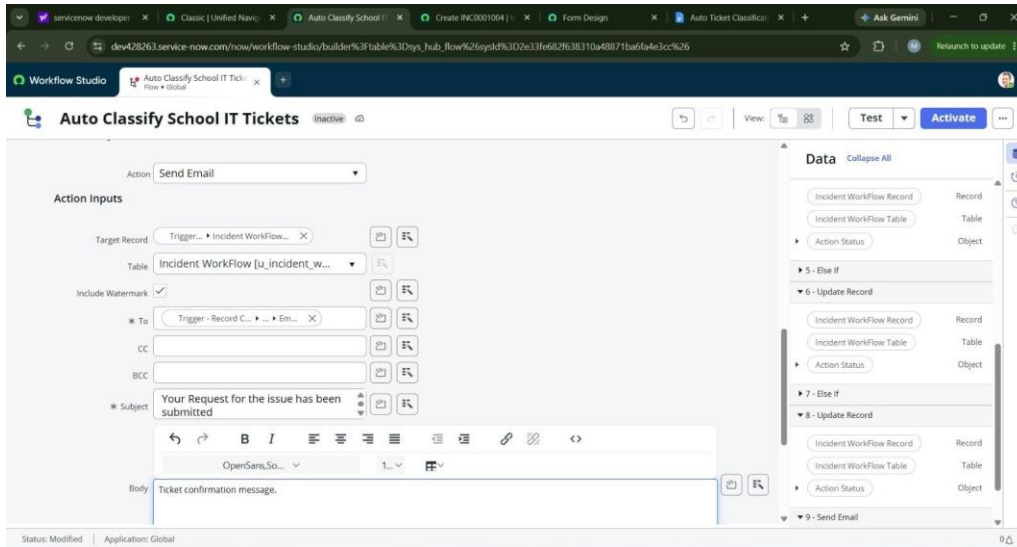
Activity : 6 Slow Computer Classification.

- Now Again Click the **Else If** Condition.
- Condition label : **SD is Performance**.
- Condition : **Short description contains Slow Computer**.
- Then Click the **Done** button.
- Now Action is an **Update Record**.
- Set **Category** = **Performance**.
- Set **Subcategory** = **Slow computer**.
- Then Click the **Done** button.



Activity : 7 Email Action Setup.

- After classification, an email notification is sent using Flow Designer.
- Set Action **Send Email**.
- To : **Caller → Email**.
- Subject : **Your Request for the issue has been submitted.**
- Body : **Ticket confirmation message.**
- Then Click the **Done** button.



- Now Click the **Activate** button from the top right corner.

PHASE 4 : Testing, Validation & Security.

- The objective of Phase 4 is to ensure that the developed solution works as expected, meets all functional requirements, and follows basic security and access control best practices.
- This phase validates the end-to-end automation, confirms data accuracy, and ensures that only authorized users can access or modify ticket data.

Data Validation Checks

- Mandatory fields captured correctly.
- Auto-number generated without duplication.
- Category and Subcategory values stored accurately.
- Reference fields resolve correctly to user and group tables.

Activity : 1 Testing & Validation.

Test Scenario 1

- Enter one Caller Name.
- Short Description: WiFi not working in library

- Then Save the Form.
- And Reload the Form.
- Then it fills auto as :
- Category → **Network**.
- Subcategory → **WiFi**.
- Email sent to caller Email Id.

Incident WorkFlow
INC00511

Number: INC00511

* Caller: Alfonso Grigen

Category: Network

Subcategory: Wi-Fi

* Short description: My college Has issue with Network

Description:

State: -- None --

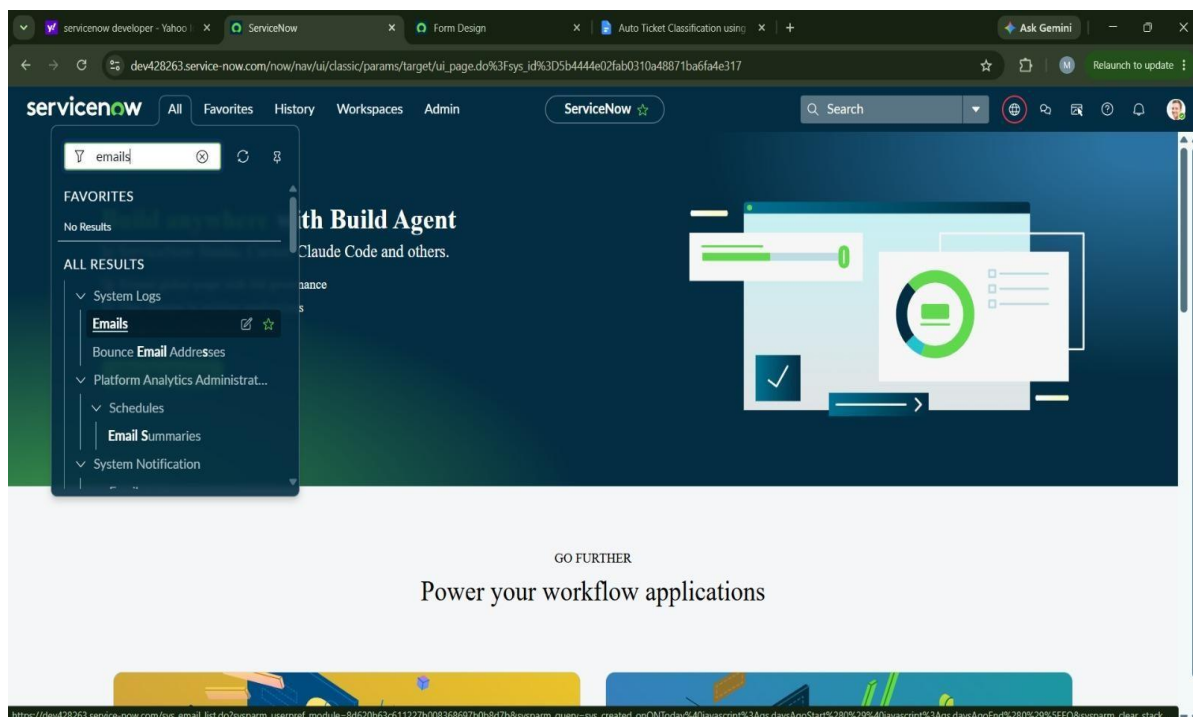
Assigned group:

Assigned to:

Update Delete

Activity : 2 Check Email Notification.

- Open All menus → Search Emails.
- Under System logs → Select Emails.
- Search with the Given Subject Name.
- Open that Record and Under Related Links Click Preview mail Option.



Activity : 3 Test Scenario 2

- Enter one Caller Name.
- Short Description: Projector not turning on.
- Then Save the Form.
- And Reload the Form.
- Then it fills auto as :
- Category → **Hardware**.
- Subcategory → **Projector**.
- Then Email sent to caller Mail id.

Incident Workflow
INC00512

Number: INC00512

* Caller: Sam Sorokin

Category: Hardware

Subcategory: Projector

* Short description: Projector not turning on

Description:

State: -- None --

Assigned group:

Assigned to:

Update Delete

PHASE 5: Deployment & Final Documentation.

- The objective of Phase 4 is to deploy the developed solution to the target environment and finalize all project documentation.
- This phase ensures that the solution is stable, reusable, and ready for demonstration or production use, along with complete and clear documentation for future reference.

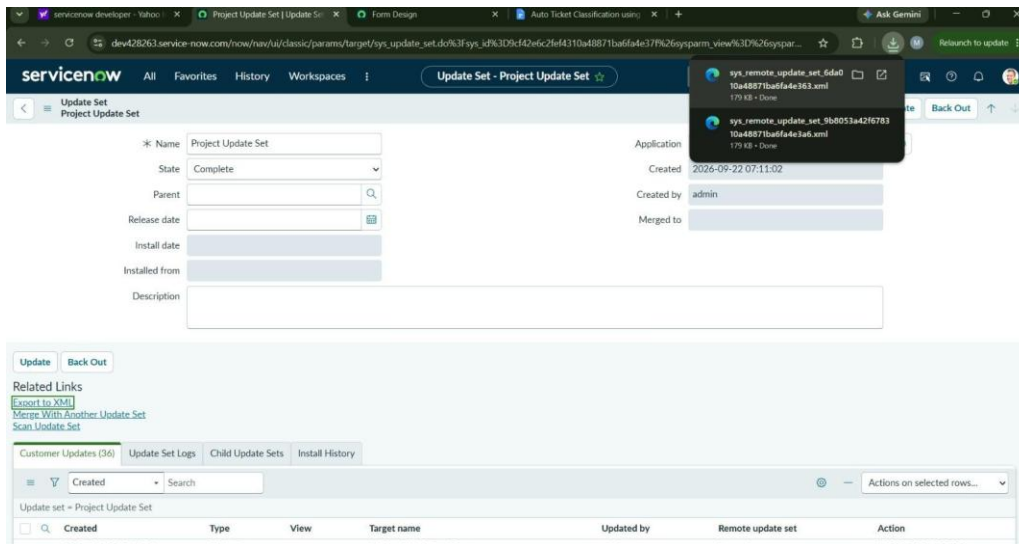
Activity : 1 Make Update set to Complete State.

- Go to All menus → Search For **Local Update Sets**.
- Under **System Update sets** Select → **Local Update sets**.
- Open the **Project Update set**.
- Make the State From **In progress** to **Complete**.
- Then **Save** the Form.
- Under Related links Click **Export to XML** Option.
- Then the Xml Format the Update set data will be **downloaded** to your system.
- We can share through WhatsApp or Mail that downloaded file.
- Then the Created Update set will transform to others in this way.

The screenshot displays the ServiceNow interface for editing an 'Update Set - Project Update Set'. The form includes the following fields and values:

- Name:** Project Update Set
- State:** Complete
- Parent:** (empty field with search icon)
- Release date:** (empty field with calendar icon)
- Install date:** (empty field)
- Installed from:** (empty field)
- Description:** (empty text area)
- Application:** Global
- Created:** 2026-09-22 07:11:02
- Created by:** admin
- Merged to:** (empty field)

Below the form, there are buttons for 'Update' and 'Back Out'. A 'Related Links' section contains links for 'Export to XML', 'Merge With Another Update Set', and 'Scan Update Set'. At the bottom, a table titled 'Customer Updates (36)' is visible, with columns for 'Created', 'Type', 'View', 'Target name', 'Updated by', 'Remote update set', and 'Action'.



Phase 6: Future Enhancement:

1. **AI Integration** – AI can be used to classify tickets more accurately.
2. **More Categories** – More ticket categories can be added for better classification.
3. **Automatic Priority** – The system can automatically identify the priority of each ticket.
4. **Email Integration** – Tickets received through email can be automatically classified.
5. **Performance Dashboard** – A dashboard can be added to track ticket status and performance.

Phase 7: Conclusion

Auto Ticket Classification using Flow Designer helps to automate the process of managing and classifying support tickets. The system identifies the ticket category and priority and automatically assigns it to the appropriate support team. This reduces manual work, saves time, and improves the efficiency of ticket management. Overall, the project provides a simple and effective solution for faster ticket routing and better customer support.

Phase 8: GitHub Repository Link

<https://github.com/NS7nevi/Auto-Ticket-Classification-using-Flow-Designer.git>

Final submission note: The student details and Public GitHub repository link have been added. The technical implementation details are based on the ServiceNow screenshots supplied for this project. Before submitting, upload this final report and the required project evidence/files to the Public GitHub repository, and enter the final submission date.

Submission note: The technical details in this document have been aligned with the ServiceNow screenshots provided for the completed implementation. The screenshots are included as project evidence.

